# Role of AI in Environmental Monitoring and Assessment

Instructor: Prof. Dr. Saman Shahid

# Agenda

- Problem Statement
- Introduction
  - What is Environmental Monitoring and Assessment?
  - What is AI?
- Role of AI in Environmental Monitoring and Assessment
  - AI in Renewable Energy
  - AI in Environmental Health
  - AI in Smart Building energy Management
  - Popular Applications of AI in Sustainable Development goals (SDGs)
- Challenges & Future Directions

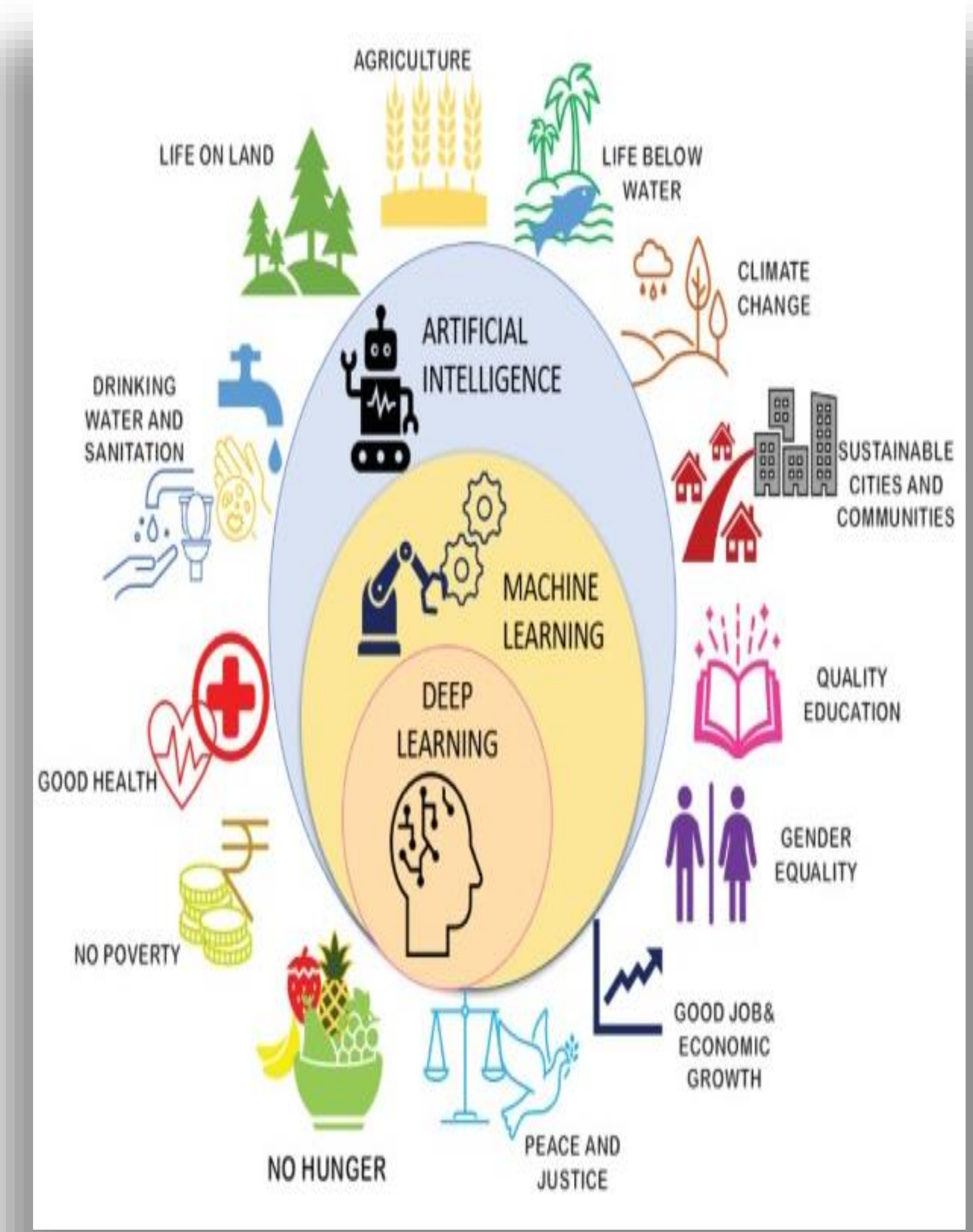
# INTRODUCTION

## What is Environmental Monitoring and Assessment?

- Environmental Monitoring and Assessment refers to the processes, tools, and techniques designed to observe an environment, characterize its quality, and establish environmental parameters.

- Environmental monitoring includes monitoring of air quality, soils, and water quality.

- Assessment, on the other hand, involves analyzing the data gathered from monitoring to accurately quantify the impact an activity has on an environment.

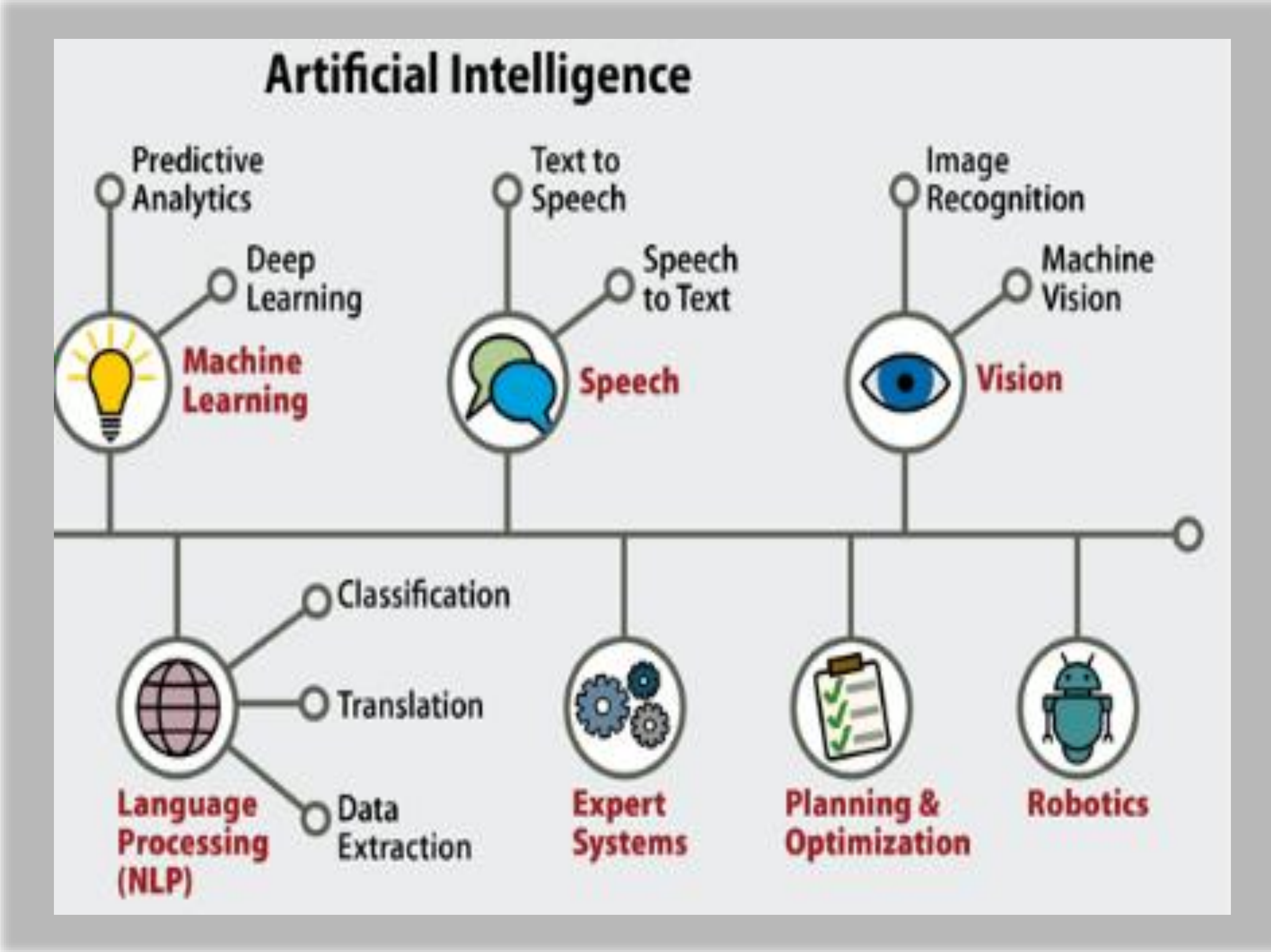


## What is Artificial Intelligence (AI)?

- Artificial Intelligence (AI) systems can perform tasks that typically require human intelligence and they can improve themselves over time based on the information they collect.

- The emergence of artificial intelligence (AI) and deep learning (DL) technologies has marked a transformative period in various sectors, particularly in sustainability.

- These innovative tools have demonstrated remarkable potential in promoting sustainable practices, especially within the realms of renewable energy and environmental health.

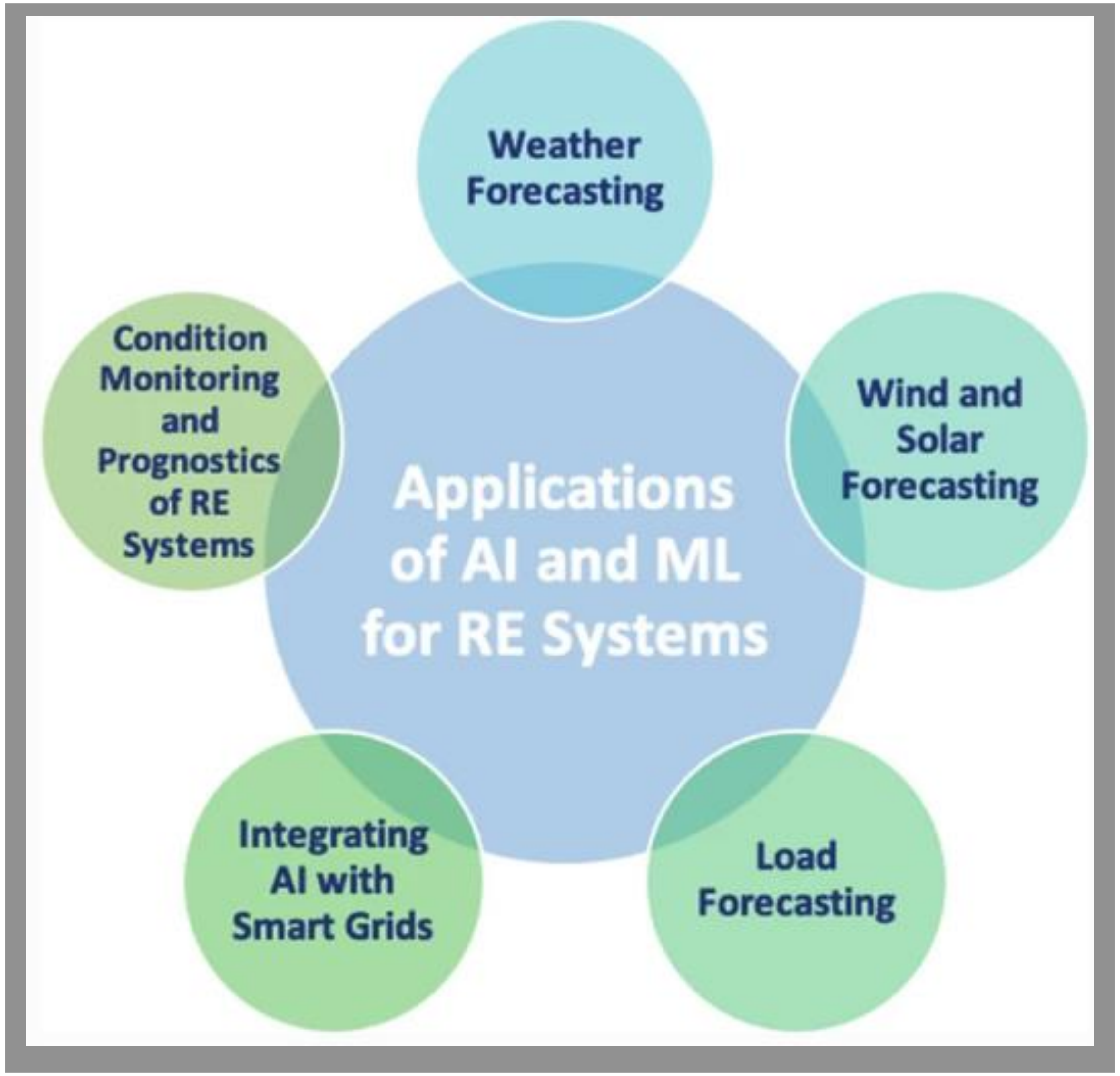


# Problem Statement

- In today's rapidly changing world, understanding and protecting our environment is more crucial than ever.
- Traditional monitoring methods often fall short, limited by cost, manpower, and the sheer vastness of data.
- This is where Artificial Intelligence (AI) emerges as a game-changer, offering powerful tools to monitor and assess the environment with unprecedented detail and accuracy.
- In this lecture, we'll explore how AI is transforming environmental science, empowering us to see the Earth like never before.

# ROLE OF AI IN ENVIRONMENTAL MONITORING AND ASSESSMENT

# AI and DL in Renewable Energy



# Literature Renewable Energy

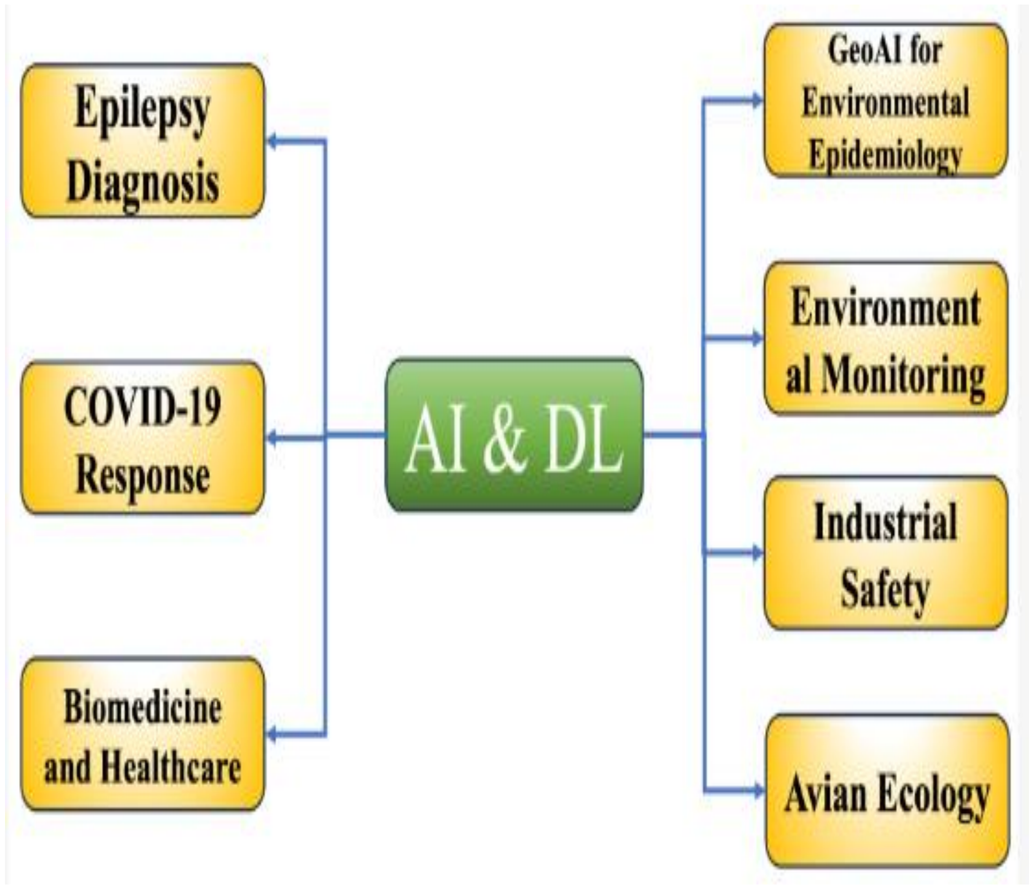
| Author Name | Year | Research Work in Renewable Energy |
|---|---|---|
| Fioretto et al. | 2019 | Combined deep learning and Lagrangian dual methods for predicting AC optimal power flows, improving the accuracy of OPF linear DC approximation |
| Zhou et al. | 2021 | Developed a deep reinforcement learning-based, data-driven method for rapid optimal AC power flow solutions, aiding power grid operators in making swift decisions |
| Moncada et al. | 2018 | Applied deep learning to forecast solar irradiance, contributing to meaningful insights for solar power optimization |
| Zhou et al. | 2021 | Proposed a hybrid deep learning method for photovoltaic (PV) energy generation forecasting, overcoming challenges with clustering techniques, CNN, LSTM, and attention mechanism |
| Gu and Li | 2022 | Provided a comprehensive review on the potential of deep learning for managing and optimizing wind and wave energy, reviewing various models and methods |
| Wang et al. | 2023 | Proposed a GM-based energy management strategy for hydrogen fuel cell buses, enhancing output efficiency and fuel economy |
| Fayyazi et al. | 2023 | Highlighted the role of AI and machine learning in the energy management, control, and optimization of hydrogen fuel cell vehicles . |
| El Jery et al. | 2023 | Utilized artificial neural networks to forecast energy efficiency and hydrogen production rates in solar thermochemical plants under various scenarios |
| Zhang et al. | 2023 | Applied a half-power prediction strategy to manage hydrogen consumption in fuel cell city buses, showcasing potential for reducing consumption and enhancing fuel economy |

# Applications of AI and DL in Environmental Health

- Traditional Machine Learning was very famous until 2010. In 2010, Deep Learning took its place by automatically detecting features from the data.

- In traditional Machine Learning, useful feature extraction was a very difficult task.

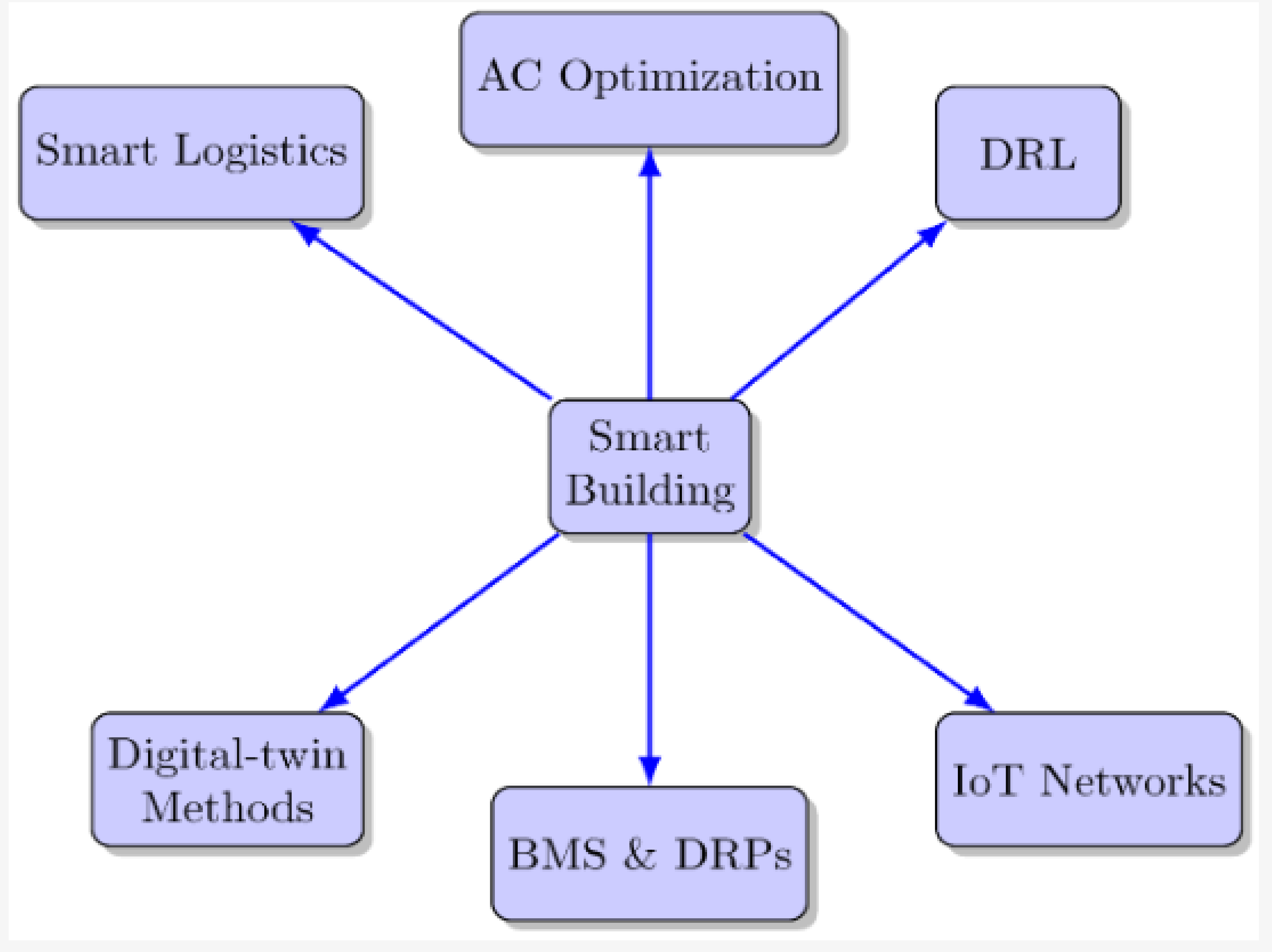
- Using Deep Learning, there are many other applications like, classification of disease, weather prediction, fraud detection in banking, early earthquake prediction, land-sliding detection and many more.



# Research Work in Environmental Health

| Author Name | Year | Research Work in Environmental Health |
|---|---|---|
| Shoeibi et al. | 2020 | Automated epileptic seizure detection using DL techniques for feature extraction from EEG and MRI data in the context of epilepsy |
| VoPham et al. | 2018 | Explored the transformative potential of geoAI in environmental epidemiology, utilizing advancements in spatial science, AI, and DL for processing large-scale spatial data |
| Alafif et al. | 2020 | Applied AI-based ML and DL methodologies for diagnosing and treating COVID-19, spanning non-invasive detection measures, severity score assessments, disease transmission modeling, and drug production |
| Lee and Lee | 2018 | Leveraged DL models for predicting harmful algal blooms in South Korea's major rivers, outperforming traditional regression analysis |
| Cirillo et al. | 2020 | Examined sex and gender gaps in biomedical technologies related to Precision Medicine, providing recommendations for optimizing utilization |
| Campero-Jurado et al. | 2020 | Presented a smart helmet prototype for industrial safety, monitoring environmental conditions in workers' environments and performing near real-time risk evaluation using AI |
| Akcay et al. | 2020 | Employed generic object-detection algorithms for avian ecology, detecting birds in natural scenes for population movement trend analysis |
| Wang and Ye | 2020 | Focused on high-quality tourism development in the Taihu Lake Basin, China, offering insights into the future of healthcare where AI-driven solutions redefine precision global health |
| Krittanawong and Kaplin | 2022 | Discussed the transformative potential of AI in global health, particularly during the COVID-19 pandemic, highlighting the capabilities of advanced DL models, such as GPT-3, in assisting physicians and redefining precision global health |

# AI and DL in Smart Building Energy Management



# Research Work in Energy Management

| Author Name | Year | Research Work in Smart Building Energy Management |
|---|---|---|
| Elsisi et al. | 2021 | Developed an innovative method combining deep learning with IoTs for air conditioner operations in smart buildings, reducing energy consumption |
| Yu et al. | 2021 | Conducted a thorough review of DRL applications in smart building energy management, addressing existing hurdles and suggesting prospective research directions |
| Agostinelli et al. | 2021 | Explored the use of digital-twin-based methods for intelligent optimization and automation systems in residential district energy management, combining 3D data models with IoTs, AI, and machine learning |
| Farzaneh et al. | 2021 | Presented a comprehensive exploration of AI applications in smart buildings, covering domains like energy, comfort, design, and maintenance, analyzed through the prism of a building management system (BMS) and demand response programs (DRPs) |
| Han et al. | 2021 | Demonstrated efficiency in managing energy within IoTs networks using robust deep-learning frameworks, employing preprocessing techniques for electricity data and an efficient decision-making algorithm for short-term forecasting |
| Andronie et al. | 2021 | Explored the interconnectedness between IoTs-based real-time production logistics and cyber-physical process monitoring systems, emphasizing the crucial role in managing the complexity and flexibility of smart factories |
| Ardabili et al. | 2022 | Critically examined recent advancements in machine learning and deep learning for building energy, introducing promising models and evaluating their performance in various forecasting tasks related to energy |
| Woschank et al. | 2020 | Provided a conceptual framework for the use of AI, machine learning, and DL in Smart Logistics, offering implications for future investigations |

# Popular Applications of AI in Sustainable Development goals (SDGs)

| SDG Goal | Application |
|---|---|
| No Poverty | Predicting poverty regions , Optimizing social security payments ,Improving microfinance services. |
| Zero Hunger | Crop yield prediction, Crop disease detection, Precision agriculture. |
| Good Health and Well-being | Disease outbreak prediction , Telemedicine services , AI-assisted diagnosis. |
| Quality Education | Personalized education ,AI-assisted grading . |
| Gender Equality | Analysis of gender bias data. |
| Clean Water and Sanitation | Water quality monitoring , Water scarcity prediction , Optimizing water distribution systems . |
| Decent Work and Economic Growth | Enhancing productivity, Job creation, Forecasting economic trends, |
| Industry, Innovation, and Infrastructure | Optimizing operations , Reducing maintenance costs ,Predictive maintenance , |
| Reduced Inequalities | Identifying and predicting social inequality ,AI in policy-making , Targeted interventions for inequality reduction |

## Artificial Intelligence for Climate Change Risk Prediction, Adaptation, & Mitigation

By developing effective models for weather forecasting and environmental monitoring, AI makes us better understand the impacts of climate change across various geographical locations.

It interprets climatic data and predicts weather events, extreme climate conditions, and other socio-economic impacts of climate change and precipitation. From a technical perspective, AI offers better climatic predictions, shows the impacts of extreme weather, finds the actual source of carbon emitters and includes numerous other reasonable contributions.

This enables the policymakers to be aware of the rising sea levels, earth hazards, hurricanes, temperature change, disruption to natural habitats, and species extinction. Nevertheless, the research community and experts have already started focusing on climate informatics with AI paradigms.

The predictive models are more appropriate to short-term forecasting models and diverge from the long-term prediction, assessment, and mitigation.  To get the maximum benefit from AI for climate change mitigation, more advanced researches are necessitated towards this domain.

# Five ways AI can combat climate change

Scientists are using AI and machine learning to fight climate change. Let’s explore a few ways this technology can reduce greenhouse gas emissions and help countries adapt to the consequences of a warmer planet.

**Transportation:** Planes, trains, automobiles, and ships constitute one of the world’s most polluting industries. However, AI can make the sector cleaner and safer. AI-enabled vehicles can minimize energy use by identifying the most efficient routes. Additionally, such technology can reduce accidents by communicating hazards between vehicles. Self-driving cars are already on the road, and fully autonomous vehicles that require no human involvement whatsoever may not be far away.

**Weather forecasting:** Meteorologists are never 100 percent accurate. But with AI, scientists can identify weather changes, heavy rainfall, and tropical cyclones with up to 99 percent accuracy. With timelier and more precise predictions, people can better prepare for storms and minimize risks—whether by evacuating or putting up storm barriers. Such technology can also anticipate wildfires, heat waves, and other extreme weather events with greater accuracy.

**Smart Agriculture:** Farming accounts globally for around 70 percent of all freshwater use. Within this sector, 40 percent of water is lost to poor resource management by some estimates. Many sprinkler heads, for instance, waste around half the water they use. But with AI technology, farmers can reduce such waste by irrigating crops more efficiently—a critical improvement as climate change exacerbates global water scarcity. AI can also help farmers optimize fertilizer use and better schedule planting seasons, leading to more productive harvests. In India, for example, AI-equipped peanut farmers have seen 30 percent higher yields from their fields.

**Urban planning:** Harnessing vast amounts of data, AI can make cities more livable and efficient amid a changing climate. AI can support disaster response, for instance, by quickly processing information and directing resources to where they are most needed following extreme weather events. AI-supported “smart cities” can also reduce resource waste by optimizing water and energy use, with such technology already rolling out in fast-growing cities in places such as Brazil and the Philippines.

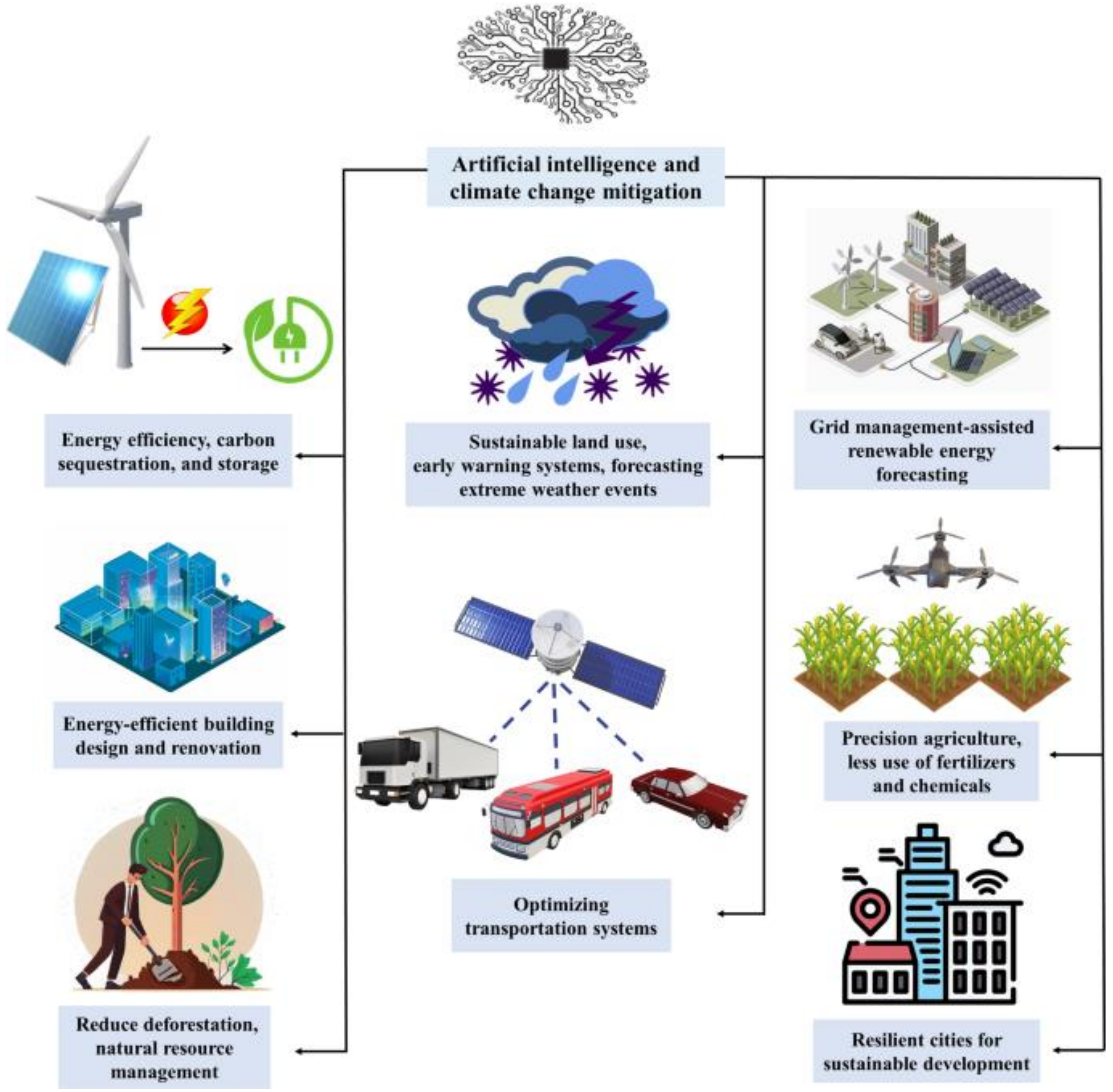
**Conservation:** Machines have long struggled with analyzing images; just think of how the most seemingly simple CAPTCHA can stump a computer. But AI is getting better at processing visual information, which is helping environmental protection efforts. AI can analyze satellite imagery far faster than humans, and this expanded processing power can quickly provide important conclusions, like whether a country’s coral reefs are dying off or if sections of forest are under threat. With greater access to information, governments can more easily monitor other countries’ environmental protection efforts and consumers can better understand how their purchases contribute to climate change.

**Smart farming agricultural technology and smart arm robots harvesting hydroponics vegetables.**

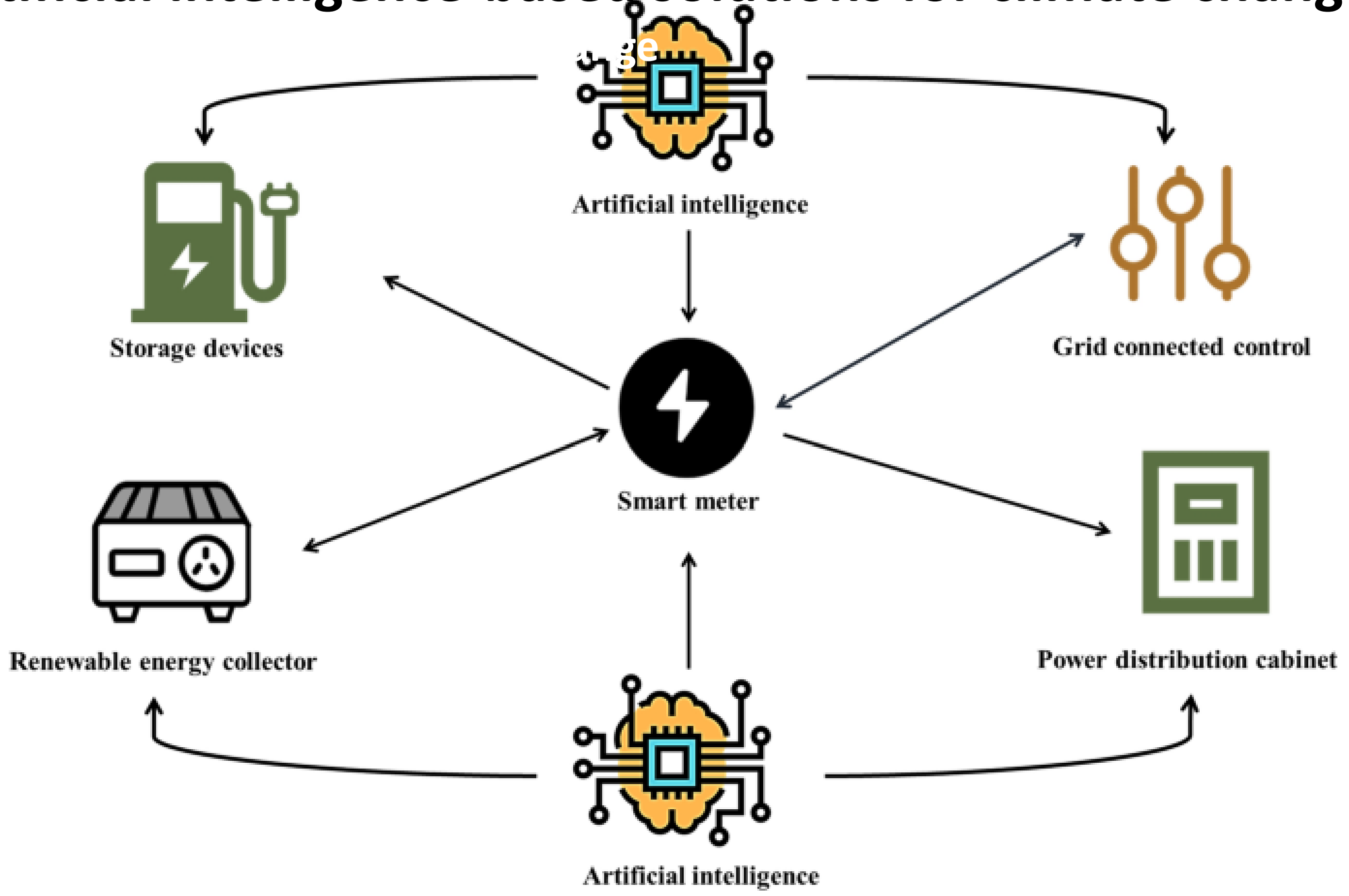
**Healthy vegetation sits alongside a field scorched by fire in the Amazon rainforest in Rondonia state, Brazil.**

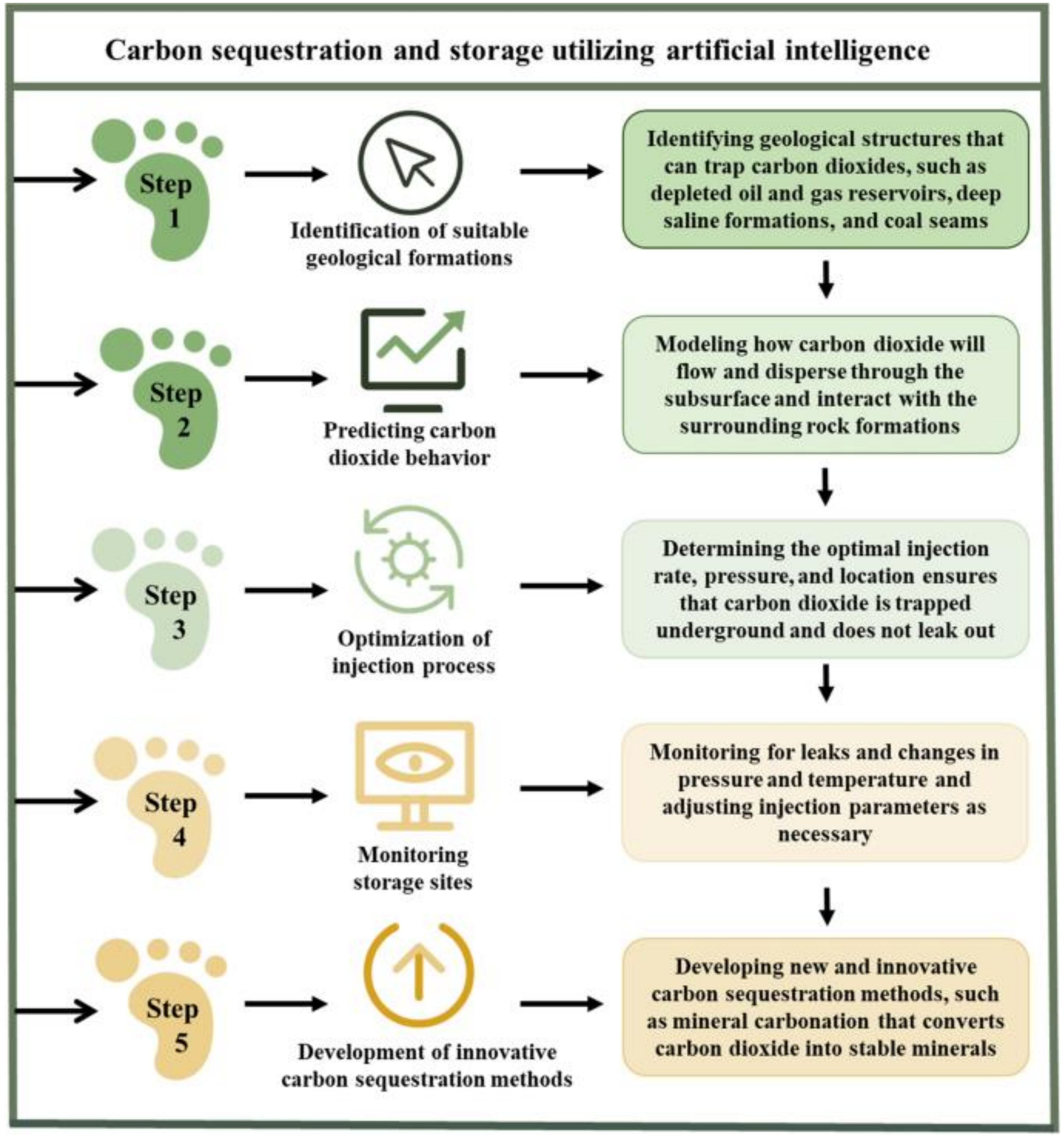
| AI technique | Adaptation-relevant examples | Description |
|---|---|---|
| Supervised learning | Wildlife surveys for ecosystem change, infrastructure monitoring, predictive maintenance | Supervised learning can help provide meaningful annotation to raw data streams. It can turn camera trap images into species population estimates, and elevation maps with weather forecasts into flood risks. |
| Transfer learning | Urban planning, early warnings for floods, crop yield monitoring | Transfer learning repurposes existing models to new contexts, often by reusing parameter estimates across tasks. This supports annotation of raw data when prior labels are scarce. For example, it can be used to tailor a model from one urban center to another. |
| Reinforcement learning and multiarmed bandits | Relocation | Reinforcement learning and multi-armed bandits estimate the long-run values of possible actions to learn optimal policies in heterogeneous, evolving environments. For example, they can be used to identify personalized incentives for relocation. |
| Semi-supervised Learning | Crop yield monitoring | Semi-supervised learning leverages unlabeled data to supplement training on small, labeled datasets. Deep network classification or Gaussian process regression come with semi-supervised variants. |
| Multimodal learning | Multifaceted impact of extreme heat, drought and food supply, resilient infrastructure planning | Multimodal techniques make it possible to integrate information across multiple modalities, like images and text. For example, they can help combine social media feeds with environmental sensor data in disaster management. |
| Multiobjective techniques | Agricultural and water management, response to heat island effects | Multiobjective methods optimize several competing objectives simultaneously, either through introduction of constraints or by reweighting goals. They ensure that model-guided improvements in one aspect of a system do not result in unintended losses elsewhere. |

So-Min Cheong, Kris Sankaran, Hamsa Bastani




## Artificial intelligence-based solutions for climate change




Lin Chen, Zhonghao Chen

- AI assisted prediction models for climate change mitigation
- Role of machine vision in climate informatics and forecasting
- Recent trends in AI to reduce carbon footprints for a sustainable environment
- AI for earth hazard management
- AI to promote eco-friendly energy production and consumption
- AI assisted expert systems for climate change risk prediction and assessment
- AI assisted big data analytics Synergy of IoT, big data, cloud computing, and AI t climate change prediction and mitigation
- Machine learning for sustainable green future
- AI in reducing the impacts of global warming
- Deep learning for sustainable earth surveillance and earth informatics

Qin Xin, Ravi Samikannu, and Chuliang Wei

# What is environmental Monitoring and its importance?

- Environmental monitoring refers to the systematic collection and analysis of data to assess the condition and trends of the natural world. This encompasses various elements, including air quality, water quality, land use, biodiversity.

**Importance:**

- **Understanding environmental trends:** Identifying and tracking changes in ecosystems over time allows us to gauge the health of the planet and understand the impacts of human activities.
- **Early warning of threats:** Detecting pollutants, resource depletion, or endangered species helps us address issues before they escalate, minimizing damage and facilitating interventions.
- **Public awareness:** Monitoring data can be used to raise awareness about environmental challenges and inform public understanding of their role in solutions.

# Challenges in Environmental Monitoring

- **Limited coverage**: Traditional monitoring stations are often sparse, leaving large areas unmonitored.
- **High cost**: Deploying and maintaining physical sensors can be expensive and resource-intensive.
- **Data overload**: The sheer volume of data collected can be overwhelming, making it difficult to extract meaningful insights.
- **Slow response times**: Traditional analysis methods can be slow, hindering timely decision-making.

# AI in Tackling Environmental Challenges

- The environmental challenges we face today are complex and urgent, demanding innovative solutions across various sectors.
- AI with its remarkable capabilities in data analysis, prediction, and optimization, emerges as a powerful tool in our fight for a sustainable future.

**Climate Change Mitigation:**

  - AI-powered smart grids can optimize energy distribution, reducing wastage and promoting renewable energy integration.
  - AI can identify optimal locations for storing captured carbon dioxide underground, preventing its release into the atmosphere.
  - AI models can predict future greenhouse gas emissions based on various factors, informing policy decisions and emission reduction strategies.

# AI in Tackling Environmental Challenges

**Environmental Monitoring and Protection:**

- AI analyzes data from sensor networks to detect and track air and water pollution in real-time, facilitating swift interventions.
- AI analyzes satellite imagery to identify deforestation activities and predict future risks, supporting conservation efforts.
- AI-powered camera traps and acoustic sensors track animal populations and behaviors, aiding in species protection and habitat management[4].

**Sustainability and Resource Management:**

- AI optimizes fertilizer and water usage in agriculture, minimizing environmental impact and maximizing crop yields.
- AI can analyze waste composition and predict waste generation, improving waste collection and recycling efficiency.
- AI facilitates the development of circular economy models by optimizing resource reuse and minimizing waste generation throughout product lifecycles.

# Case Studies

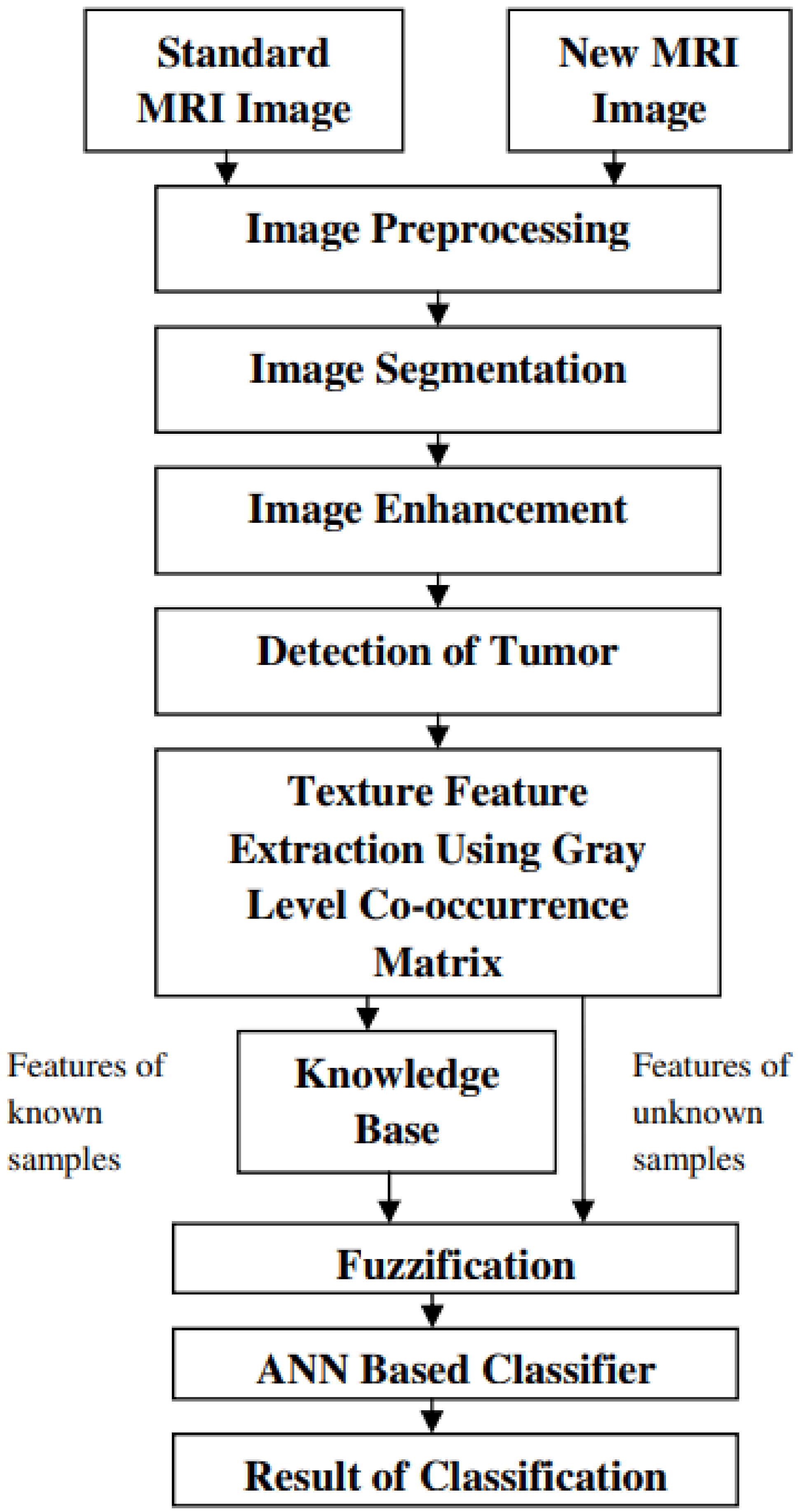
## UNEP's World Environment Situation Room (WESR)

- The United Nations Environment Program (UNEP's) WESR is a dynamic online platform that provides access to real-time, interactive data and information on the environment.
- It is a resource for policymakers, scientists, journalists, and the general public to track environmental trends and make informed decisions.
- The WESR uses a variety of data sources, including satellite imagery, sensor networks, and social media, to provide a comprehensive picture of the environment.
- The platform also includes tools for analyzing data, creating visualizations, and sharing information with others.
- The WESR has been used to track a number of environmental issues, including:
  - Air quality
  - Water quality
  - Land use
  - Climate change
  - Biodiversity loss
- The WESR is a valuable tool for understanding the state of the environment and making informed decisions about its future.

# Case Studies

## International Methane Emissions Observatory (IMEO)

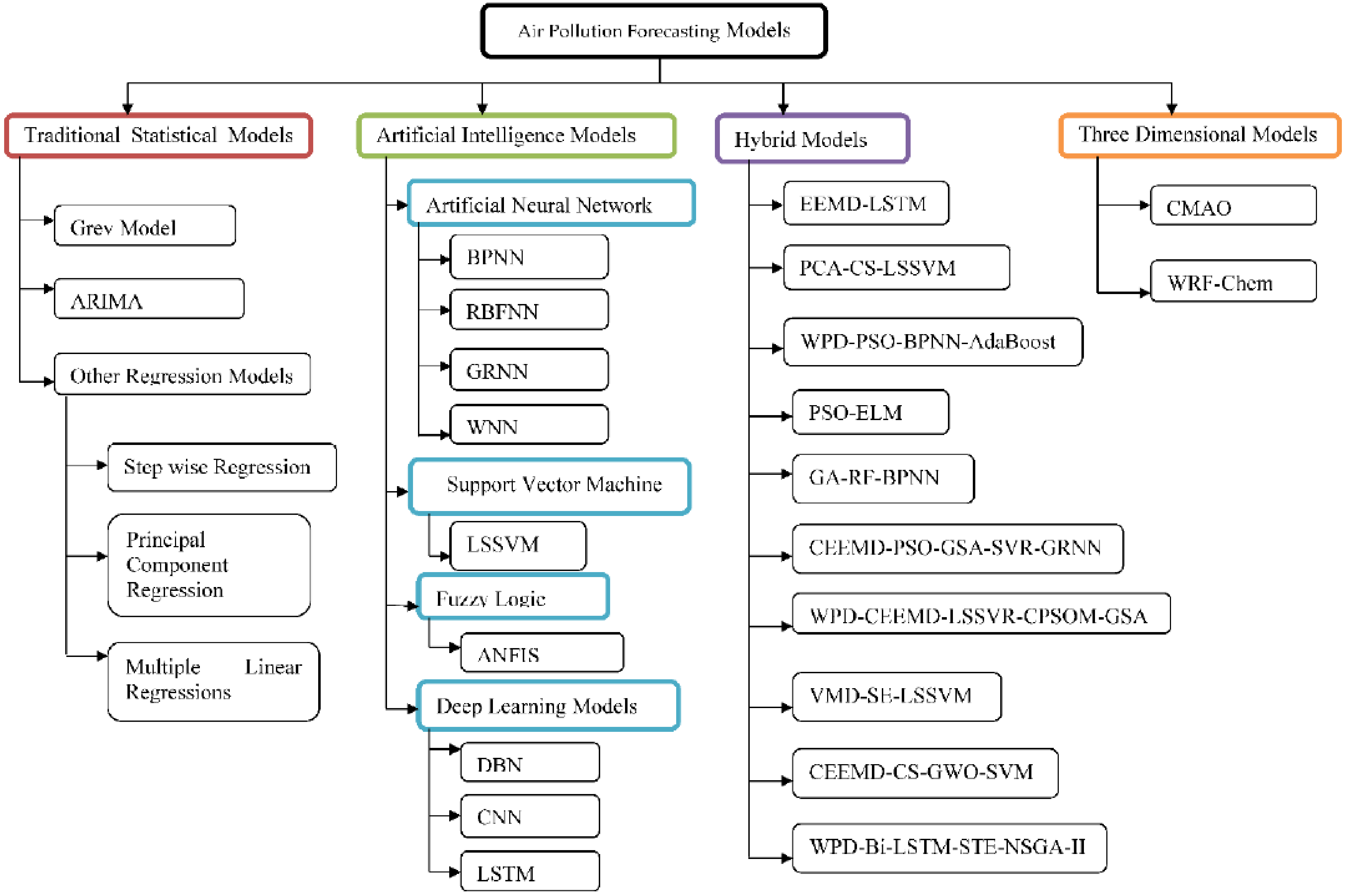
- IMEO is a global initiative that was launched in 2020 by the UNEP to improve the measurement, reporting, and verification (MRV) of methane emissions.
- Methane is a potent greenhouse gas that is 80 times more effective at trapping heat than carbon dioxide over a 20-year period.
- The IMEO works with a variety of partners, including governments, industry, and research institutions, to develop and implement new MRV methods for methane emissions.
- The IMEO also provides training and capacity building to help countries improve their methane MRV capabilities.
- The IMEO has a number of objectives, including:
- To improve the accuracy and completeness of methane emissions data
- To increase the transparency of methane emissions reporting
- To promote the development and implementation of new methane mitigation technologies
- To help countries meet their commitments under the Paris Agreement
- The IMEO is a vital tool for reducing methane emissions and combatting climate change.




# Classification of Brain Cancer Using Artificial Neural Network

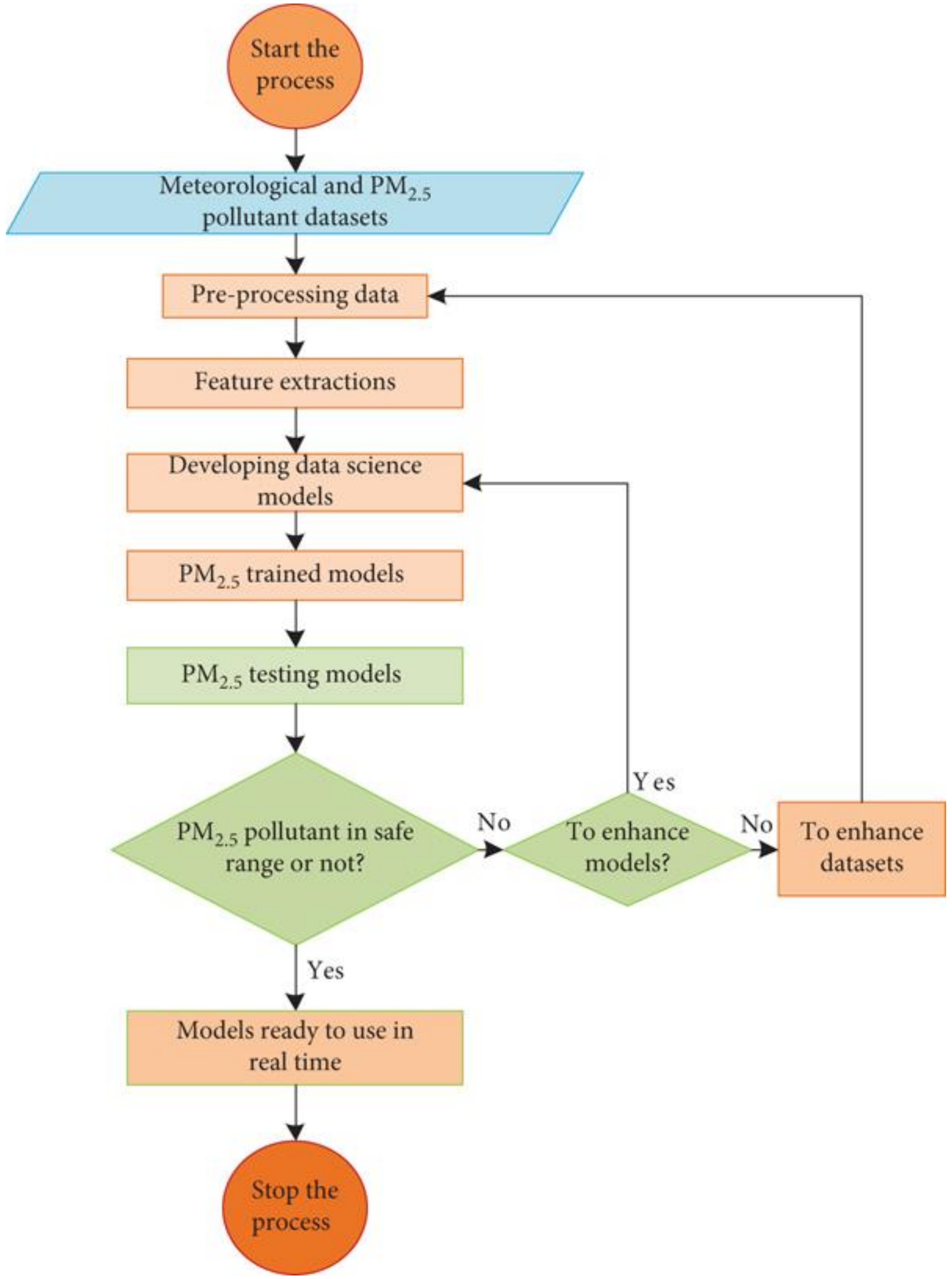
Dipali M. Joshi, Dr.N. K. Rana

# Different Methods of Air Pollution Forecasting.




Shankar Subramaniam

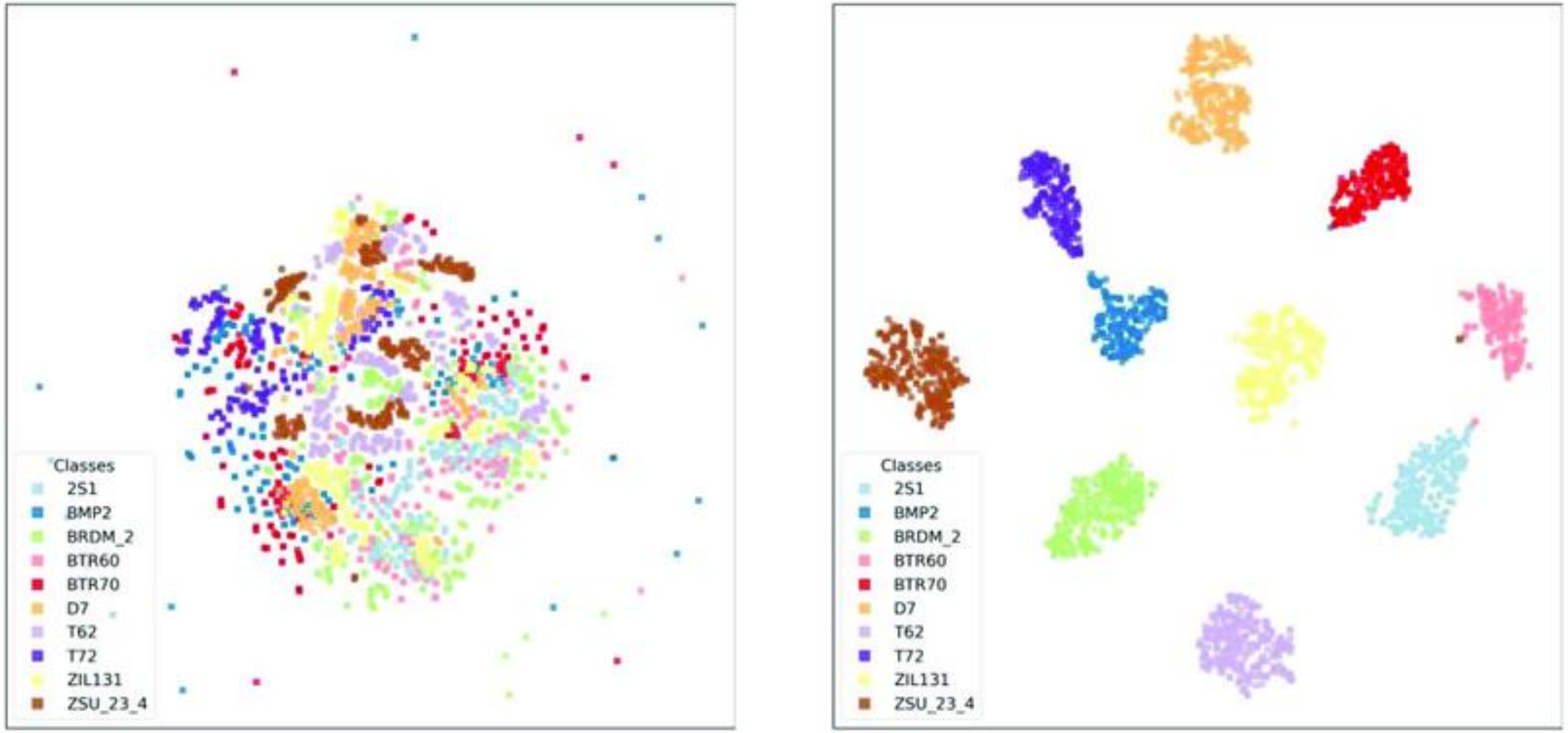
# Flowchart representations for predicting $PM_{2.5}$ and air quality forecasting.




**Kothandaraman**

# Simple Application

- In Image (a), the data is very dense and each by visualization.
- By utilizing AI techniques our goals is to separate the distribution of these spaces.




(a) (b)

## Challenges

- Lack of transparency in AI and DL models
- Scalability and high dimensionality of data
- Integration of AI and DL with 6G networks
- Ethics and privacy concerns
- Energy efficiency in AI and DL models
- Environmental impact of AI systems

## Future Directions

- Develop interpretable AI models
- Develop scalable AI and DL algorithms
- Employ distributed computing and parallel processing
- Investigate data compression techniques
- Develop robust bias detection and mitigation techniques
- Investigate model compression, pruning, and quantization

# Summary

# Summary

- This lecture underscores the extensive impact of AI and DL in monitoring environmental assessment across sectors such as SDGs, renewable energy, environmental health, and smart building energy management.

- AI holds the potential to address 134 out of 169 SDG targets, regulatory oversight is crucial to ensure transparency and ethical standards.

- The demonstrated potential in optimizing energy management and enhancing diagnostic procedures in renewable energy and environmental health is notable.

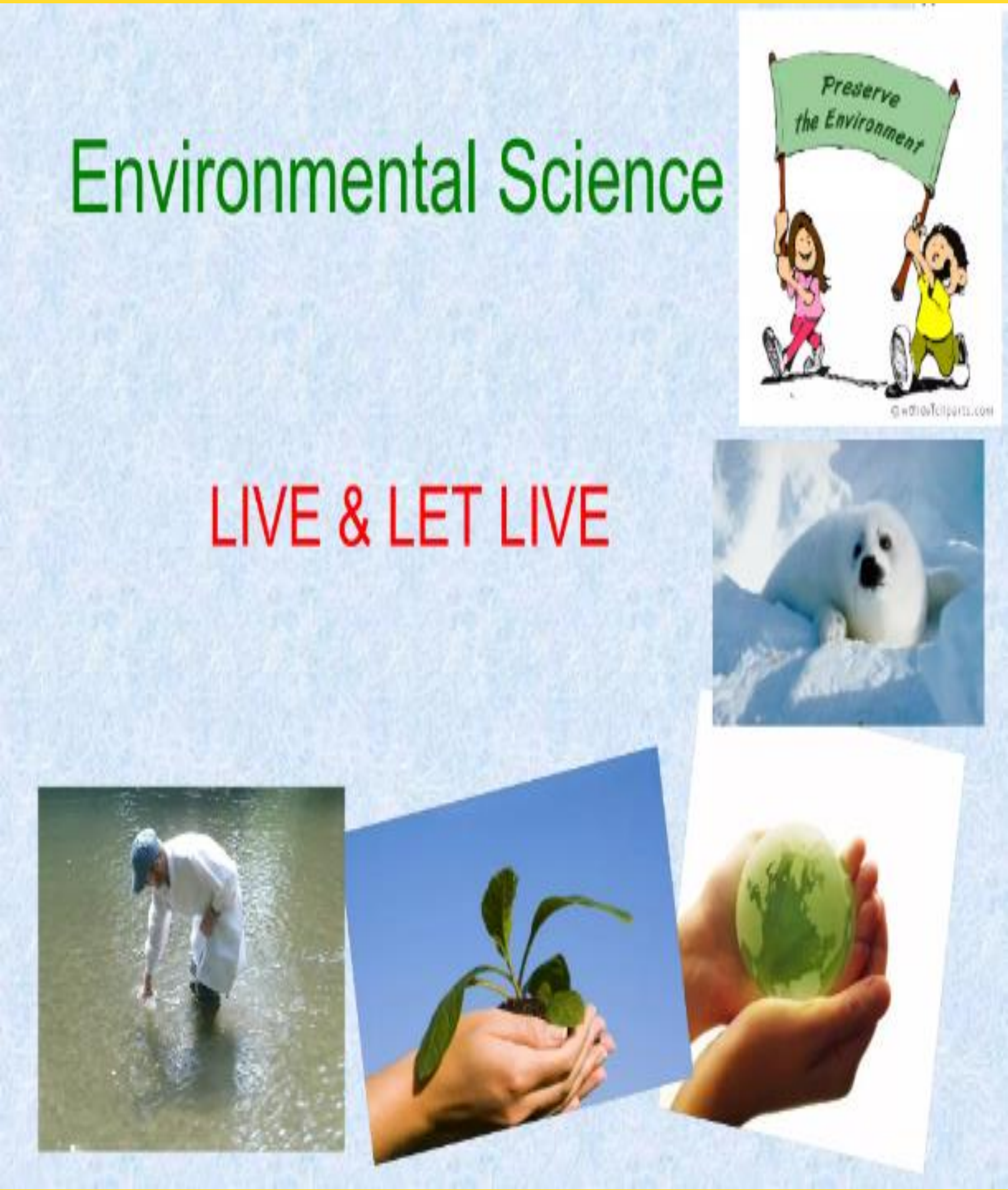
- However, challenges like explainability, scalability, and ethical considerations must be addressed.

- Future directions should prioritize enhancing transparency, scalability, integrating with advanced networks, and improving energy efficiency to unlock the full potential of AI and DL for a sustainable future.

# References

1. Fioretto, F.; Mak, T.W.; Van Hentenryck, P. Predicting ac optimal power flows: Combining deep learning and lagrangian dual methods. In Proceedings of the AAAI Conference on Artificial Intelligence; Association for the Advancement of Artificial Intelligence: Washington, DC, USA, 2020; Volume 34, pp. 630–637. Zhou, Y.; Zhang, B.; Xu, C.; Lan, T.; Diao, R.; Shi, D.; Wang, Z.; Lee, W.J. A data-driven method for fast ac optimal power flow solutions via deep reinforcement learning. J. Mod. Power Syst. Clean Energy 2020, 8, 1128–1139. [CrossRef]
2. Moncada, A.; Richardson, W., Jr.; Vega-Avila, R. Deep learning to forecast solar irradiance using a six-month UTSA skyimager dataset. Energies 2018, 11, 1988. [CrossRef]
3. Zhou, H.; Liu, Q.; Yan, K.; Du, Y. Deep learning enhanced solar energy forecasting with AI-driven IoT. Wirel. Commun. Mob. Comput. 2021, 2021, 1–11. [CrossRef]
4. Mahmoud, M.; Slama, S.B. Peer-to-Peer Energy Trading Case Study Using an AI-Powered Community Energy Management System. Appl. Sci. 2023, 13, 7838. [CrossRef]
5. Fayyazi, M.; Sardar, P.; Thomas, S.I.; Daghigh, R.; Jamali, A.; Esch, T.; Kemper, H.; Langari, R.; Khayyam, H. Artificial Intelligence/Machine Learning in Energy Management Systems, Control, and Optimization of Hydrogen Fuel Cell Vehicles. Sustainability 2023, 15, 5249. [CrossRef]
6. El Jery, A.; Salman, H.M.; Al-Khafaji, R.M.; Nassar, M.F.; Sillanpää, M. Thermodynamics Investigation and Artificial Neural Network Prediction of Energy, Exergy, and Hydrogen Production from a Solar Thermochemical Plant Using a Polymer Membrane Electrolyzer. Molecules 2023, 28, 2649. [CrossRef] [PubMed]
7. Zhang, L.; Ning, L.; Yang, X.; Zeng, S.; Yuan, T.; Li, G.; Ke, C.; Zhang, J. Half-Power Prediction and Its Application on the Energy Management Strategy for Fuel Cell City Bus. Automot. Innov. 2023, 6, 131–142. [CrossRef]
8. Shoeibi, A.; Khodatars, M.; Ghassemi, N.; Jafari, M.; Moridian, P.; Alizadehsani, R.; Panahiazar, M.; Khozeimeh, F.; Zare, A.; Hosseini-Nejad, H.; et al. Epileptic seizures detection using deep learning techniques: A review. Int. J. Environ. Res. Public Health 2021, 18, 5780. [CrossRef] [PubMed]
9. VoPham, T.; Hart, J.E.; Laden, F.; Chiang, Y.Y. Emerging trends in geospatial artificial intelligence (geoAI): potential applications for environmental epidemiology. Environ. Health 2018, 17, 1–6. [CrossRef]
10. Alafif, T.; Tehame, A.M.; Bajaba, S.; Barnawi, A.; Zia, S. Machine and deep learning towards COVID-19 diagnosis and treatment: survey, challenges, and future directions. Int. J. Environ. Res. Public Health 2021, 18, 1117. [CrossRef]

11. Lee, S.; Lee, D. Improved prediction of harmful algal blooms in four Major South Korea's Rivers using deep learning models. Int. J. Environ. Res. Public Health 2018, 15, 1322. [CrossRef]
12. Cirillo, D.; Catuara-Solarz, S.; Morey, C.; Guney, E.; Subirats, L.; Mellino, S.; Gigante, A.; Valencia, A.; Rementeria, M.J.; Chadha, A.S.; et al. Sex and gender differences and biases in artificial intelligence for biomedicine and healthcare. NPJ Digit. Med. 2020, 3, 81. [CrossRef]
13. Campero-Jurado, I.; Márquez-Sánchez, S.; Quintanar-Gómez, J.; Rodríguez, S.; Corchado, J.M. Smart helmet 5.0 for industrial internet of things using artificial intelligence. Sensors 2020, 20, 6241. [CrossRef] [PubMed]
14. Akçay, H.G.; Kabasakal, B.; Aksu, D.; Demir, N.; Öz, M.; Erdo ˘gan, A. Automated bird counting with deep learning for regional bird distribution mapping. Animals 2020, 10, 1207. [CrossRef] [PubMed]
15. Wang, G.; Ye, L. Spatial-Temporal Pattern of Mismatch Degree of High-Quality Tourism Development and Its Formation Mechanism in Taihu Lake Basin, China. Sustainability 2022, 14, 4812. [CrossRef]
16. . Krittanawong, C.; Kaplin, S. Artificial intelligence in global health. Eur. Heart J. 2021, 42, 2321–2322 . [CrossRef]
17. Woschank, M.; Rauch, E.; Zsifkovits, H. A review of further directions for artificial intelligence, machine learning, and deep learning in smart logistics. Sustainability 2020, 12, 3760. [CrossRef)
18. Elsisi, M.; Tran, M.Q.; Mahmoud, K.; Lehtonen, M.; Darwish, M.M. Deep learning-based industry 4.0 and internet of things towards effective energy management for smart buildings. Sensors 2021, 21, 1038. [CrossRef] [PubMed
19. Agostinelli, S.; Cumo, F.; Guidi, G.; Tomazzoli, C. Cyber-physical systems improving building energy management: Digital twin and artificial intelligence. Energies 2021, 14, 2338. [CrossRef]
20. Farzaneh, H.; Malehmirchegini, L.; Bejan, A.; Afolabi, T.; Mulumba, A.; Daka, P.P. Artificial intelligence evolution in smart buildings for energy efficiency. Appl. Sci. 2021, 11, 763. [CrossRef]
21. Han, T.; Muhammad, K.; Hussain, T.; Lloret, J.; Baik, S.W. An efficient deep learning framework for intelligent energy management in IoT networks. IEEE Internet Things J. 2020, 8, 3170–3179. [CrossRef]
22. Ardabili, S.; Abdolalizadeh, L.; Mako, C.; Torok, B.; Mosavi, A. Systematic review of deep learning and machine learning for building energy. Front. Energy Res. 2022, 10, 786027. [CrossRef]


Environmental Science
Preserve
the Environment
LIVE & LET LIVE

**THANK YOU!**